

Superba Tunnel Profiler

How to use — v1.0.3

This app downloads logged temperature data from a Fluke 54-II-B thermometer over its IR cable, and turns it into a time/temperature profile you can view, compare, search, and export.

No installation needed. Extract the SuperbaTunnelProfiler folder anywhere and run **SuperbaTunnelProfiler.exe** inside it — keep the whole folder together, the app stores its data (**superba_profiler.db**) right next to the .exe.

What you need

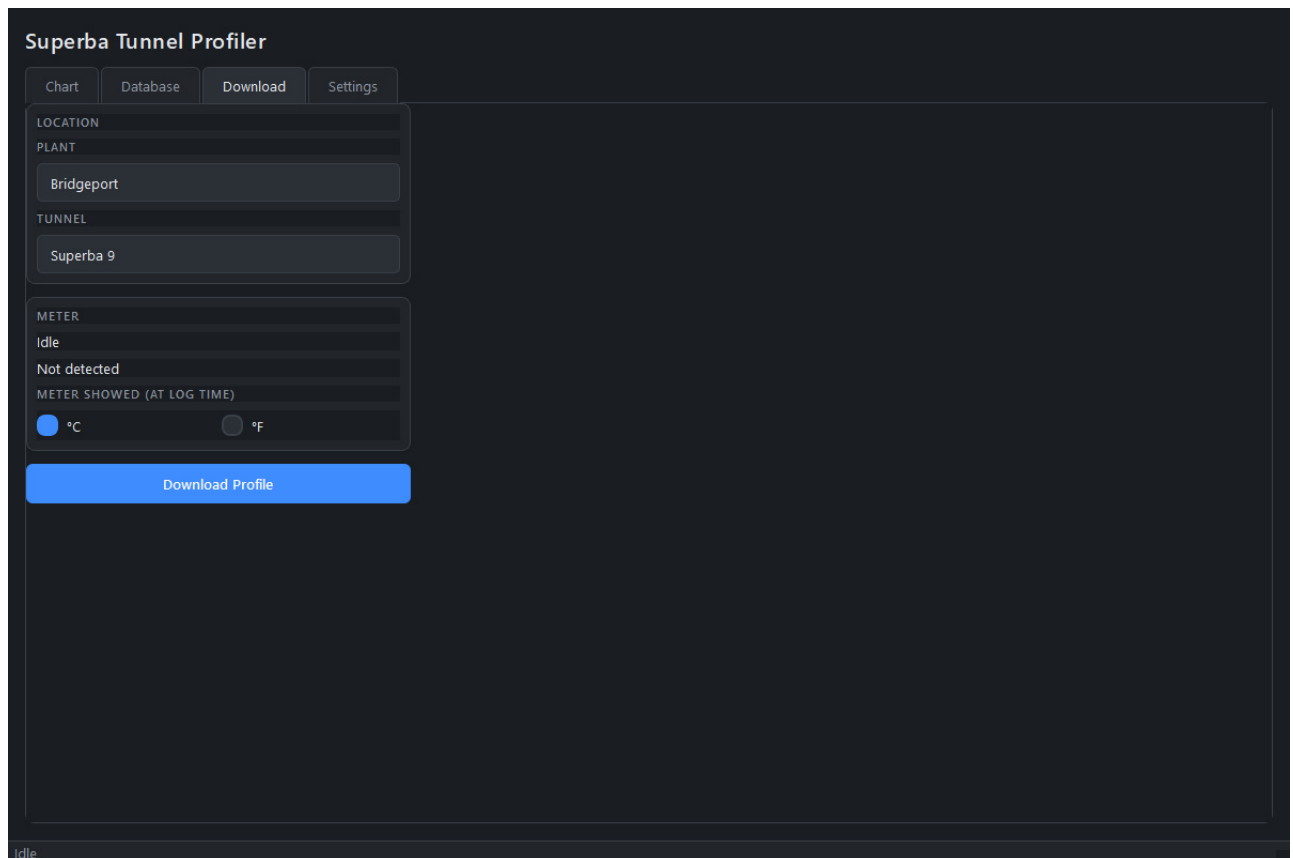
- A Fluke 54-II-B thermometer with logged readings on it
- The Fluke IRUSB Rev.-II cable, connected between the meter and your PC
- Windows — nothing else to install

The four tabs

Tab	What it's for
Chart	View, compare, and export the profiles you have plotted
Database	Search every profile ever downloaded, and open, export, or print any of them
Download	Pull a new profile off the meter
Settings	Display preferences (currently: °C / °F)

Downloading a profile

1. Connect the IRUSB cable to the meter's IR window and to your PC.
2. On the meter, press **SHIFT + RECALL** so the display shows "**Ir SEnd**". The app can only talk to the meter while it's in this mode.
3. In the app, open the **Download** tab.
4. Fill in **Plant** and **Tunnel** — free text, so use whatever naming you already use on the floor (a tunnel doesn't have to be numbered the same way at every plant).
5. Under **Meter Showed (at log time)**, select whether the meter's display was set to °C or °F when the log was recorded. This is just a record of what you saw on the meter — the app always stores and can display the data in either unit regardless.
6. Click **Download Profile**. The status line will say "Downloading..." while it works.



The Download tab: Location, Meter status, and the Download button.

If the download fails: double check the meter is actually showing "Ir SEnd" (it times out and drops back to the normal display after a while, so you may need to press SHIFT + RECALL again), and that the cable is seated in the meter's IR window, not just resting near it.

Once a download succeeds, the app switches to the **Chart** tab automatically and plots what you just pulled off the meter.

Viewing and comparing charts



The Chart tab, with two runs overlaid — the red run has been dragged 8s to the right to line up its ramp with the blue run.

Plotting more than one run

Double-click any entry in **Recent Profiles** to add it to the graph, in its own color. Double-click it again to remove just that one. This lets you overlay several runs to compare them directly — the highlighted (bold, colored) entries in the list are the ones currently on the graph.

Lining up runs that started at different times

The meter's logging start time is set by hand, so two runs of the same tunnel rarely start at exactly the same point in the process. To fix that: **click and drag directly on a curve** to slide it left or right in time (temperature is left alone) until its start and end line up with the run you're comparing it to. A small label shows the run name and current time offset while you drag.

Double-click a curve to reset its offset back to zero. Each run remembers its offset for as long as it stays on the graph, even if you switch °C/°F or temporarily remove and re-add it from Recent Profiles.

Reading the graph

- Move the mouse over the graph to see a crosshair with the exact time and temperature.
- Scroll the mouse wheel to zoom the time axis in or out.
- Hold **Ctrl** and scroll to zoom the temperature axis instead (it's capped to a fixed 0–195°C / 0–390°F range so the scale stays meaningful).

Exporting what's on the graph

Button	What it does
Export Graph as PNG	Saves an image of the graph exactly as it looks right now
Export Plotted Data (CSV)	Saves the time/temperature numbers for every run currently plotted, including any drag alignment you've applied
Export Plotted Data (Excel)	Same data, in a spreadsheet with a built-in chart already drawn
Clear Graph	Removes everything from the graph (doesn't delete anything from the database)

Searching past profiles

The screenshot shows the 'Superba Tunnel Profiler' interface with the 'Database' tab selected. The interface includes a search bar with filters for 'Plant' and 'Tunnel', and date range selectors for 'From' and 'To'. A 'Search' button is present. Below the search bar is a table with the following columns: Date, Plant, Tunnel, Peak, Points, and Logged. The table contains six rows of data, all for 'Superba A'.

Date	Plant	Tunnel	Peak	Points	Logged
2026-08-05T18:38:23+00:00		Superba A	92.7°C	31	°C
2026-08-05T18:07:02+00:00		Superba A	92.7°C	31	°C
2026-08-05T18:03:46+00:00		Superba A	36.9°C	39	°C
2026-08-05T17:46:33+00:00		Superba A	28.7°C	39	°C
2026-08-05T17:44:32+00:00		Superba A	27.4°C	31	°C
2026-08-05T17:41:58+00:00		Superba A	27.4°C	31	°C

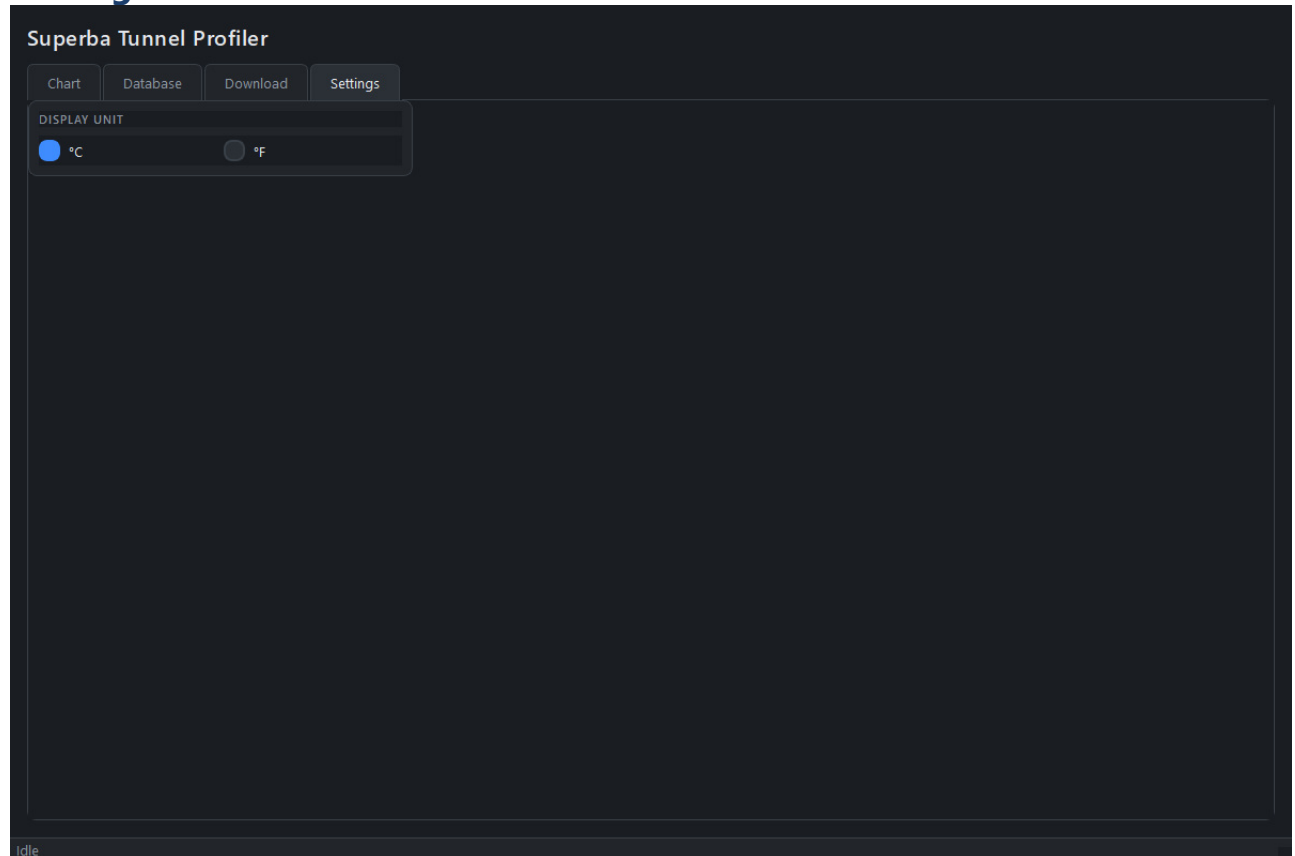
At the bottom of the table, it says '6 RUNS'. To the right of the table are two buttons: 'Export Data (CSV)' and 'Print Report'.

The Database tab: every profile ever downloaded, searchable.

Use this tab once you have more history than fits in the Chart tab's Recent Profiles list.

- Filter by **Plant** and/or **Tunnel** (matches partial text), and by a date range using the **From** / **To** checkboxes.
- **Double-click** a row to open it on the Chart tab.
- **Right-click** a row for a menu to export just that run as a PNG, CSV, or Excel file.
- Select a row and use **Export Data (CSV)** or **Print Report** at the bottom for the same options, plus a one-page printable summary with the run's stats and its graph.

Settings



The Settings tab.

Display Unit controls whether temperatures are shown in °C or °F throughout the app — on the graph, in both run lists, and in exports. It's independent of what the meter was showing when a profile was logged.

Where your data lives

Every downloaded profile is stored in **superba_profiler.db**, a single file that sits right next to **SuperbaTunnelProfiler.exe**. If you move or copy the app to a new folder or PC, take that file with it (or start a fresh one on purpose). There's no cloud sync — it's local to whichever folder the .exe is run from.

Troubleshooting

Problem	Try this
Meter port shows "Not detected"	Reconnect the IRUSB cable, and confirm Windows sees it as a COM port (Device Manager).
Download button does nothing / times out	The meter must be showing "Ir SEnd" (SHIFT + RECALL). It reverts on its own after a period of inactivity, so re-trigger it right before clicking

	Download.
"The meter returned zero readings"	The selected log on the meter is empty. Recall through the meter's stored logs to confirm there's actually data on it.
A run is missing from Recent Profiles	That list only shows the 50 most recent. Use the Database tab to search further back.
A dragged curve won't line up exactly	Zoom in on the time axis (scroll wheel) first — it's much easier to nudge a curve precisely at a higher zoom level.